

Affinity landscape of the intra-tumoral B-cells in lung adenocarcinoma

THOMAS DUPIC

Which antigens are targeted by intra-tumoral B cells?



Affinity landscape of the intra-tumoral B-cells in lung adenocarcinoma

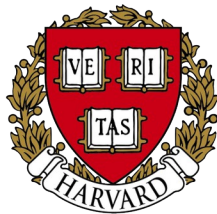
THOMAS DUPIC

Which antigens are targeted by intra-tumoral B cells?



Post-Doc
– A. Walczak et
T. Mora, Paris
*Statistics of the
immune
repertoire*
6 pub., 2 1st aut.

LPENS
LABORATOIRE DE PHYSIQUE
DE L'ÉCOLE NORMALE SUPÉRIEURE



2018

2020

2025



Affinity landscape of the intra-tumoral B-cells in lung adenocarcinoma

THOMAS DUPIC

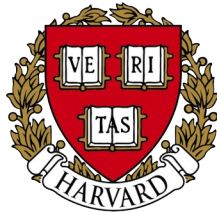
Which antigens are targeted by intra-tumoral B cells?



Post-Doc
– A. Walczak et
T. Mora, Paris
*Statistics of the
immune
repertoire*
6 pub., 2 1st aut.

Post-Doc
– M. Desai, Boston
*Affinity landscape
of proteins*
7 pub., 3 1st aut.

LPENS
LABORATOIRE DE PHYSIQUE
DE L'ÉCOLE NORMALE SUPÉRIEURE



2018

2020

2025



Affinity landscape of the intra-tumoral B-cells in lung adenocarcinoma

THOMAS DUPIC

Which antigens are targeted by intra-tumoral B cells?



Post-Doc

– A. Walczak et
T. Mora, Paris
*Statistics of the
immune
repertoire*
6 pub., 2 1st aut.

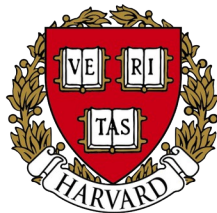
LPENS

LABORATOIRE DE PHYSIQUE
DE L'ÉCOLE NORMALE SUPÉRIEURE

2018

Post-Doc

– M. Desai, Boston
*Affinity landscape
of proteins*
7 pub., 3 1st aut.



2020

Team Leader at CIML, Marseille
*Affinities within the immune
repertoire.*

Hybrid team, experimental and
computational

Ciml
IMMUNOLOGY

2025



Affinity landscape of the intra-tumoral B-cells in lung adenocarcinoma

THOMAS DUPIC

Which antigens are targeted by intra-tumoral B cells?



Post-Doc

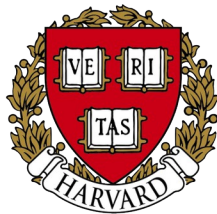
– A. Walczak et
T. Mora, Paris
*Statistics of the
immune
repertoire*
6 pub., 2 1st aut.

LPENS
LABORATOIRE DE PHYSIQUE
DE L'ÉCOLE NORMALE SUPÉRIEURE

2018

Post-Doc

– M. Desai, Boston
*Affinity landscape
of proteins*
7 pub., 3 1st aut.



2020



Team Leader at CIML, Marseille
*Affinities within the immune
repertoire.*
Hybrid team, experimental and
computational



2025

CSS 2: Cancer &
Genetic diseases

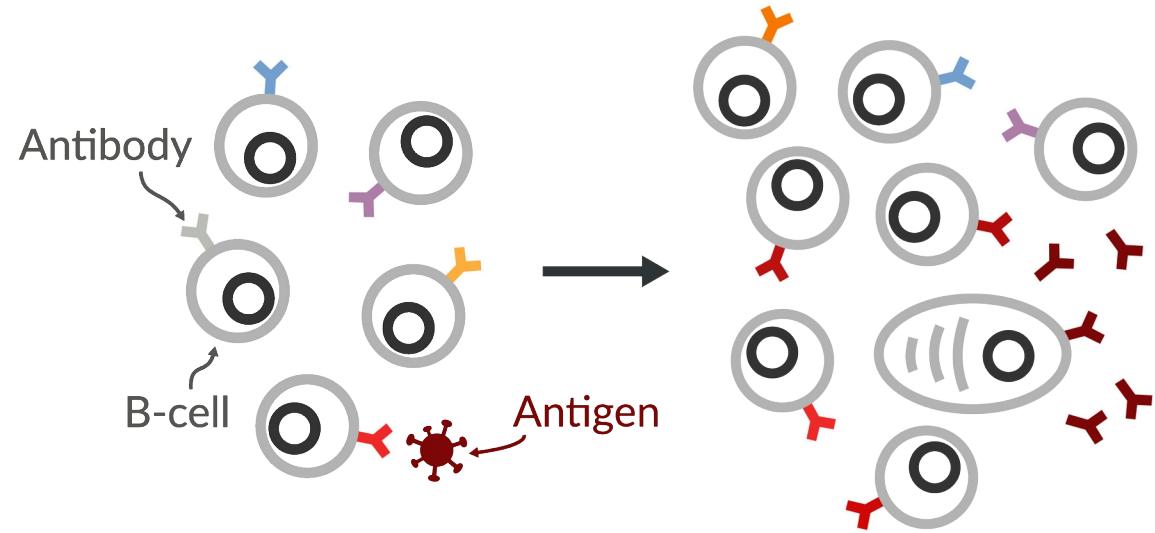
Inserm

Immune repertoires

Immune repertoire : set of all antibodies / T-cell receptors in one individual ($\approx 10^{13}$).

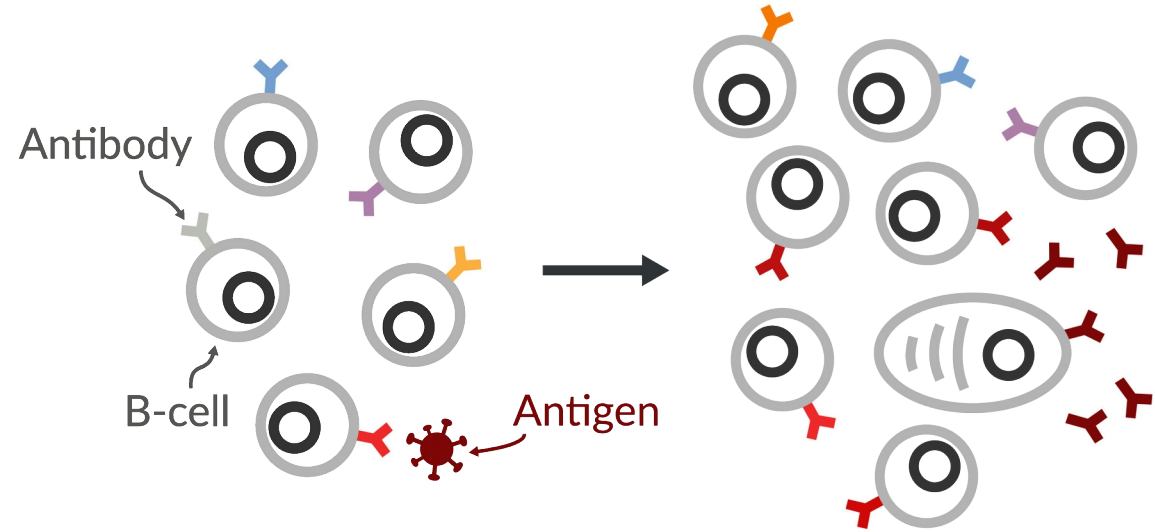
Immune repertoires

Immune repertoire : set of all antibodies / T-cell receptors in one individual ($\approx 10^{13}$).



Immune repertoires

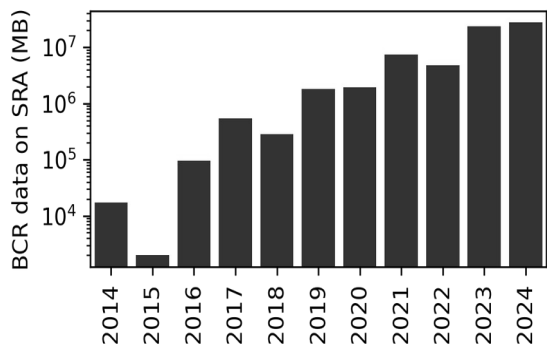
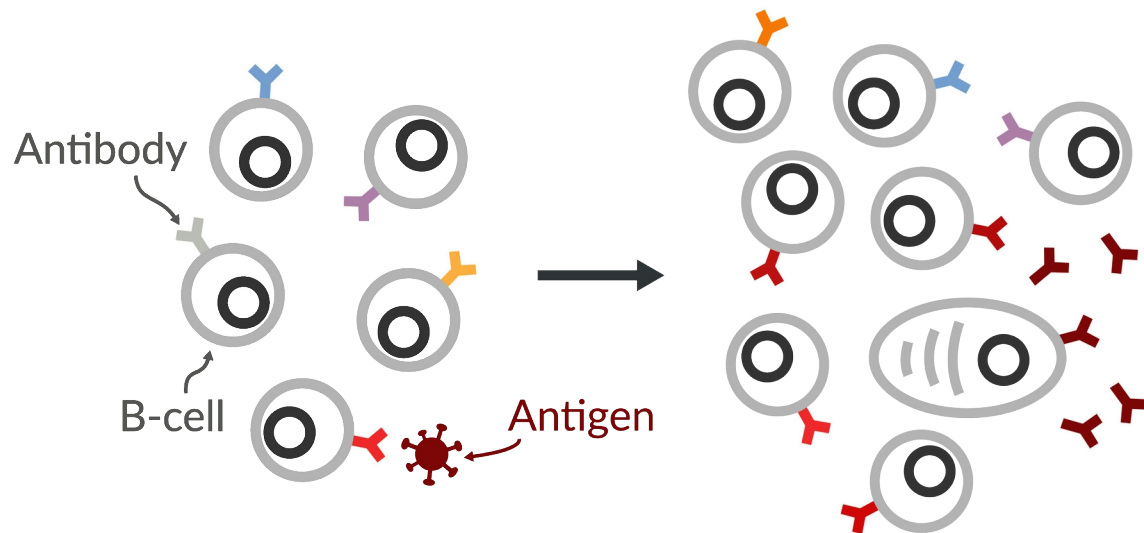
Immune repertoire : set of all antibodies / T-cell receptors in one individual ($\approx 10^{13}$).



The repertoire as a prognosis tool ?

Immune repertoires

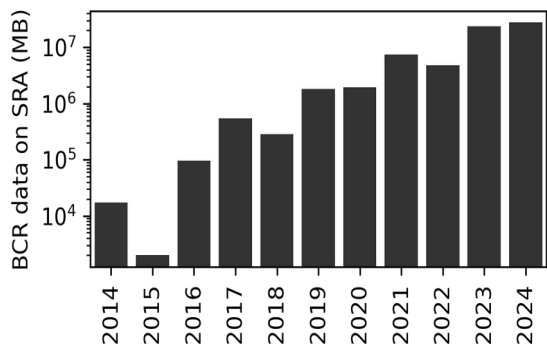
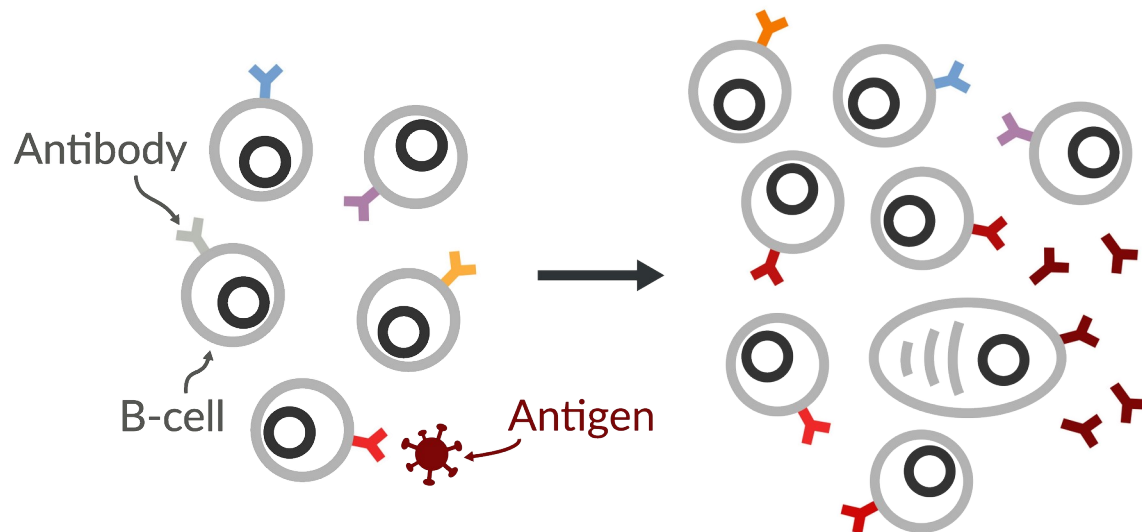
Immune repertoire : set of all antibodies / T-cell receptors in one individual ($\approx 10^{13}$).



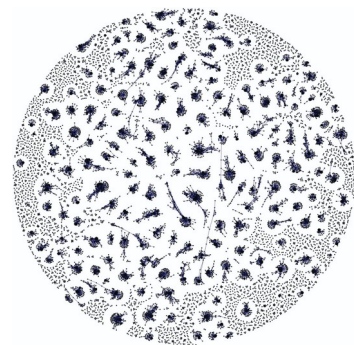
The repertoire as a prognosis tool ?
Easy to access...

Immune repertoires

Immune repertoire : set of all antibodies / T-cell receptors in one individual ($\approx 10^{13}$).



The repertoire as a prognosis tool ?
Easy to access... but hard to interpret.



Immune repertoires

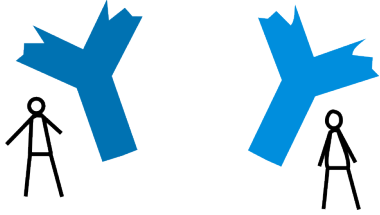
Statistical models, **generation**¹ & **selection**² of the receptors. **Dupic et al., PLOS Comp. Bio (2019)**

$$P(\text{Y}) = \dots$$

Immune repertoires

Statistical models, **generation**¹ & **selection**² of the receptors. **Dupic** et al., PLOS Comp. Bio (2019)

$$P(\text{Y-shaped receptor}) = \dots$$



Public repertoire: do shared receptors imply a shared response?

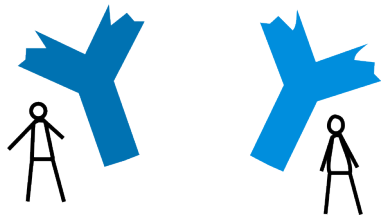
Dupic et al., PLOS Gen. (2021)
Abbate, **Dupic** et al., PNAS (2024)

Immune repertoires

Statistical models, **generation**¹ & **selection**² of the receptors.

Dupic et al., PLOS Comp. Bio (2019)

$$P(\text{Y}) = \dots$$



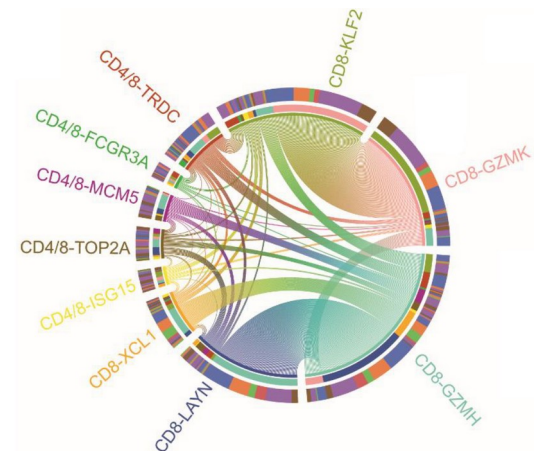
Public repertoire: do shared receptors imply a shared response?

Dupic et al., PLOS Gen. (2021)

Abbate, Dupic et al., PNAS (2024)

Tracking tumor-infiltrating lymphocytes in lung cancer

Gueguen, ..., Dupic et al., Sci. Immunol. (2021)



Immune repertoires

Statistical models, **generation**¹ & **selection**² of the receptors.

Dupic et al., PLOS Comp. Bio (2019)

$$P(\text{Y}) = \dots$$



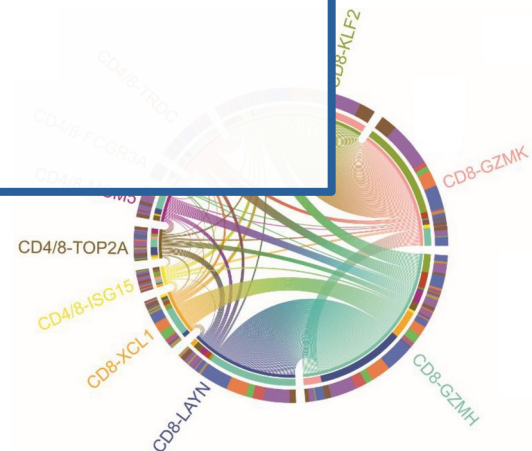
The receptor is used as a **barcode** to track a clone, to estimate diversity.

shared Response?

Dupic et al., PLOS Gen. (2021)
Abbate, Dupic et al., PNAS (2024)

Tracking tumor-infiltrating lymphocytes in lung cancer

Gueguen, ..., Dupic et al., Sci. Immunol. (2021)



Immune repertoires

Statistical models, **generation**¹ & **selection**² of the receptors.

Dupic et al., PLOS Comp. Bio (2019)

$$P(\text{Y}) = \dots$$



The receptor is used as a **barcode** to track a clone, to estimate diversity.

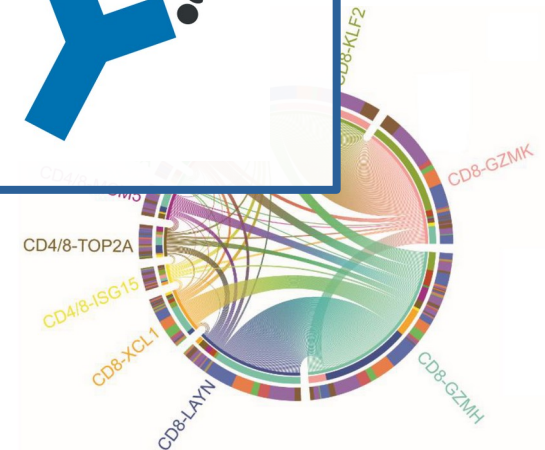
Current methods don't integrate the **functional properties** of the receptor.



Dupic et al.,
Abbate, Dupic

Tracking tumor-infiltrating lymphocytes in
lung cancer

Gueguen, ..., Dupic et al., Sci. Immunol. (2021)



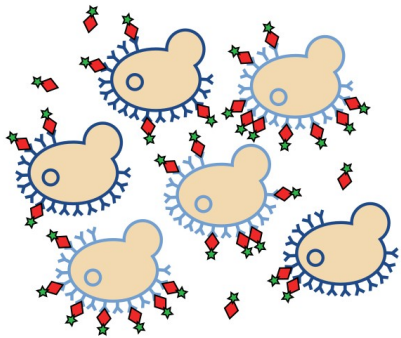
Beyond barcodes : tracking interactions between proteins in high-throughput

Tite-Seq: high-throughput and quantitative method can measure up to 100'000 proteins against one ligand.



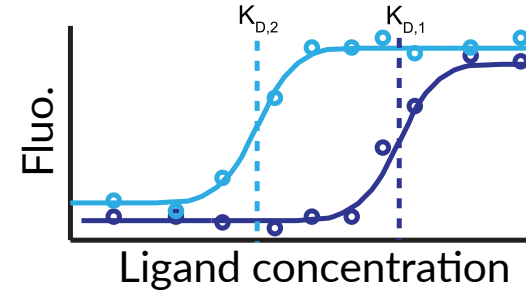
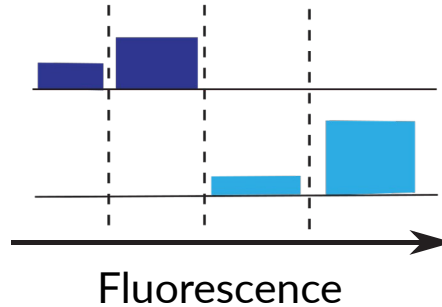
Beyond barcodes : tracking interactions between proteins in high-throughput

Tite-Seq: high-throughput and quantitative method can measure up to 100'000 proteins against one ligand.



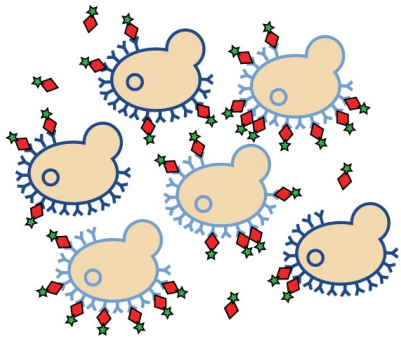
Expression on the yeast with fluorescent antigen

Cell sorting, sequencing and statistics



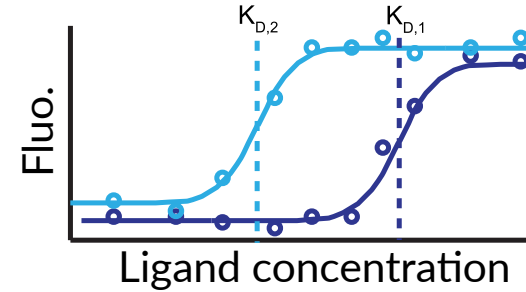
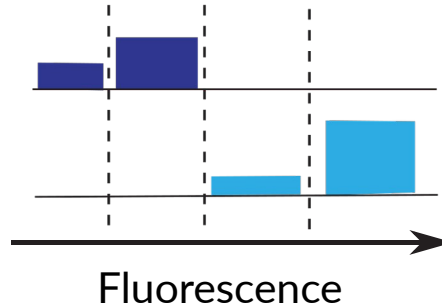
Beyond barcodes : tracking interactions between proteins in high-throughput

Tite-Seq: high-throughput and quantitative method can measure up to 100'000 proteins against one ligand.



Expression on the yeast with fluorescent antigen

Cell sorting, sequencing and statistics



Viral proteins

Moulana*, Dupic* et al., Nat. Comm. (2022)

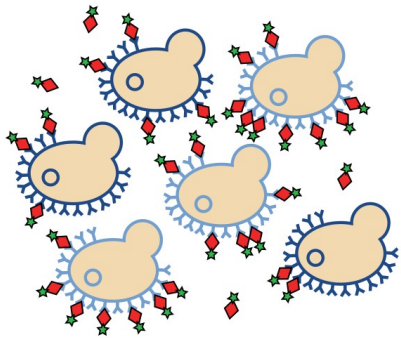
Moulana*, Dupic* et al., eLife (2023)

Moulana, Dupic et al., MBE (2026)



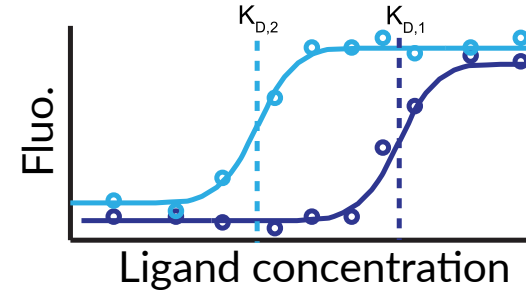
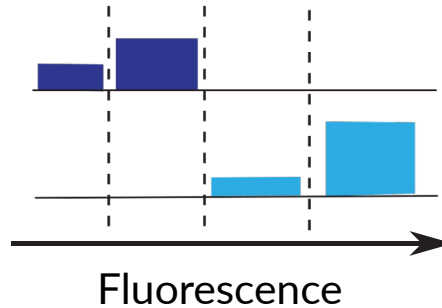
Beyond barcodes : tracking interactions between proteins in high-throughput

Tite-Seq: high-throughput and quantitative method can measure up to 100'000 proteins against one ligand.



Expression on the yeast with fluorescent antigen

Cell sorting, sequencing and statistics



Viral proteins

Moulana*, Dupic* et al., Nat. Comm. (2022)

Moulana*, Dupic* et al., eLife (2023)

Moulana, Dupic et al., MBE (2026)

Antibodies

Phillips, ..., Dupic et al., eLife (2021)

Phillips, ..., Dupic et al., eLife (2023)

Dupic, et al., Genetics (2026, in revision)



Measuring affinity at repertoire scale



Measuring affinity at repertoire scale

Long-term B-cell
response to SARS-
CoV-2 vaccination

Moulana, ..., Dupic*, Desai* (in preparation)



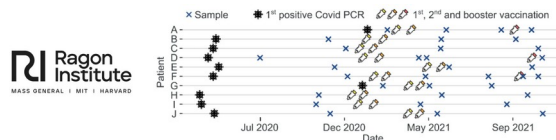
Measuring affinity at repertoire scale

Long-term B-cell
response to SARS-
CoV-2 vaccination



Measuring affinity at repertoire scale

Long-term B-cell
response to SARS-
CoV-2 vaccination



Repertoire Sequencing →



Common BCRs →

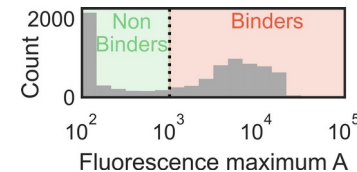


Affinity

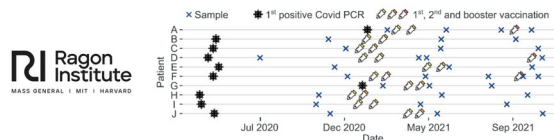
Moulana, ..., Dupic*, Desai* (in preparation)



Measuring affinity at repertoire scale



Long-term B-cell
response to SARS-
CoV-2 vaccination



1M BCRs tested $\Rightarrow$ 5000 binders



Repertoire Sequencing $\longrightarrow$



Common BCRs $\longrightarrow$

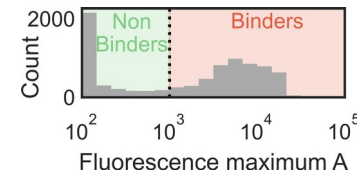


Affinity

Moulana, ..., Dupic*, Desai* (in preparation)

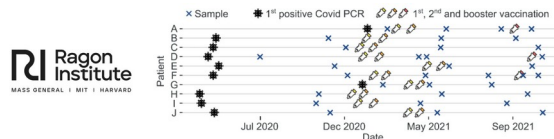


Measuring affinity at repertoire scale



1M BCRs tested $\Rightarrow$ 5000 binders

Long-term B-cell
response to SARS-
CoV-2 vaccination



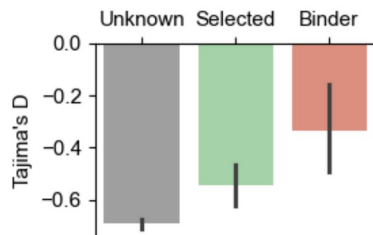
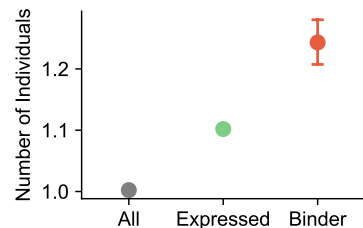
Repertoire Sequencing $\longrightarrow$



Common BCRs $\longrightarrow$



Affinity

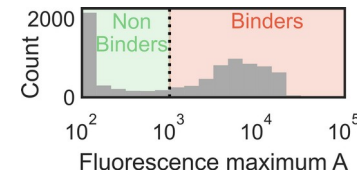


The repertoire statistics of **binders**
and **non-binders** differ.

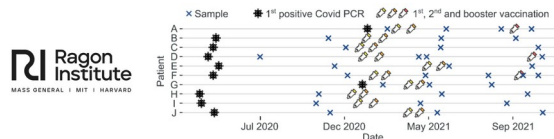
Moulana, ..., Duplic*, Desai* (in preparation)



Measuring affinity at repertoire scale



Long-term B-cell response to SARS-CoV-2 vaccination



1M BCRs tested $\Rightarrow$ 5000 binders



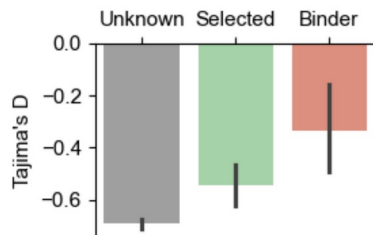
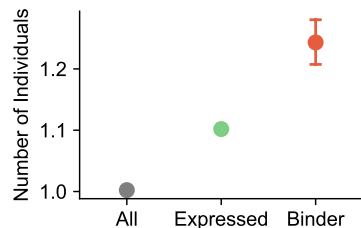
Repertoire Sequencing $\rightarrow$



Common BCRs $\rightarrow$



Affinity



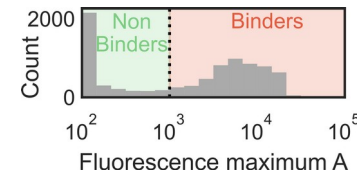
The repertoire statistics of **binders** and **non-binders** differ.

Improving binding models with transfer learning

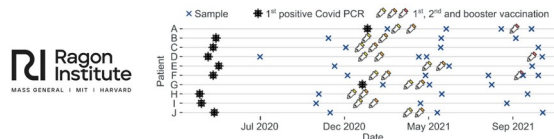
Moulana, ..., Duplic*, Desai* (in preparation)



Measuring affinity at repertoire scale



Long-term B-cell
response to SARS-
CoV-2 vaccination



1M BCRs tested $\Rightarrow$ 5000 binders



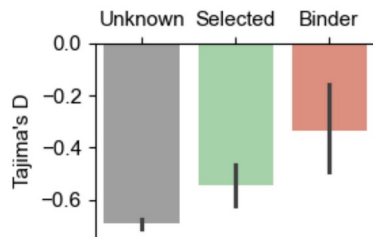
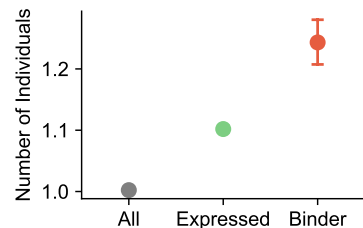
Repertoire Sequencing $\longrightarrow$



Common BCRs $\longrightarrow$



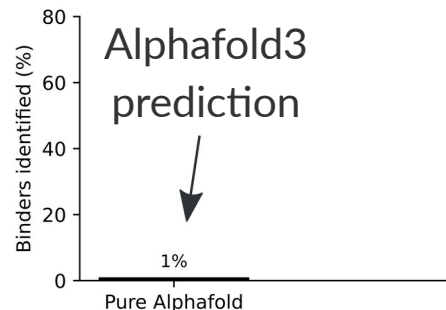
Affinity



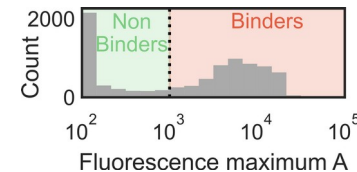
The repertoire statistics of **binders**
and **non-binders** differ.

Improving binding models
with transfer learning

Moulana, ..., Dupic*, Desai* (in preparation)

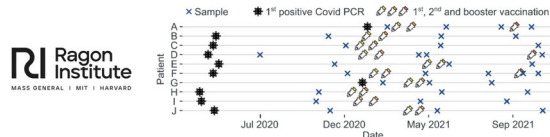


Measuring affinity at repertoire scale



1M BCRs tested $\Rightarrow$ 5000 binders

Long-term B-cell
response to SARS-
CoV-2 vaccination



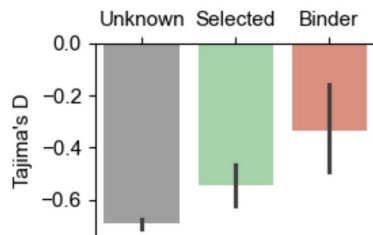
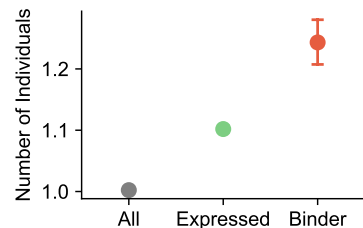
Repertoire Sequencing $\rightarrow$



Common BCRs $\rightarrow$



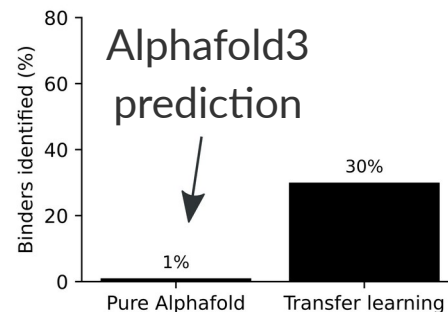
Affinity



The repertoire statistics of **binders**
and **non-binders** differ.

Improving binding models
with transfer learning

Moulana, ..., Duplic*, Desai* (in preparation)



AlphaFold3
prediction
 $\times$
Our measurements

Measuring affinity at repertoire scale



Select
antibodies of
interest

Measuring affinity at repertoire scale

Tite-Seq × repertoire :



Sequenced
repertoires

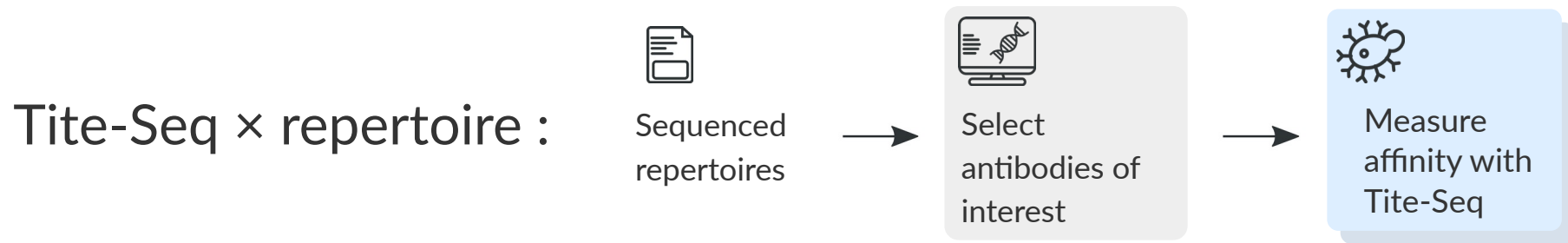


Select
antibodies of
interest



Measure
affinity with
Tite-Seq

Measuring affinity at repertoire scale



Tite-Seq is precise, mature and cheap

Measuring affinity at repertoire scale

Tite-Seq × repertoire :



Sequenced
repertoires

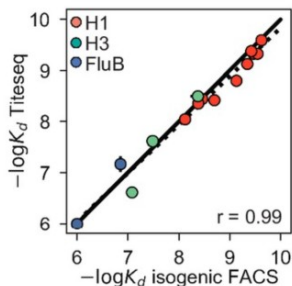


Select
antibodies of
interest



Measure
affinity with
Tite-Seq

Tite-Seq is precise, mature and cheap



Precision
comparable to
low-throughput
methodes

Measuring affinity at repertoire scale

Tite-Seq × repertoire :



Sequenced
repertoires

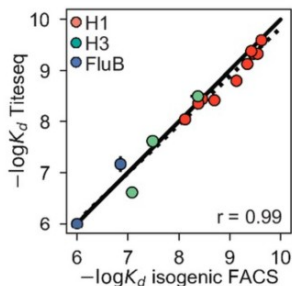


Select
antibodies of
interest



Measure
affinity with
Tite-Seq

Tite-Seq is precise, mature and cheap



Precision
comparable to
low-throughput
methodes

Starr et al., Nature, 2021
Greaney et al., Cell Host & Microbes, 2023
DeWitt, W. S. et al., Cell, 2026

Measuring affinity at repertoire scale

Tite-Seq × repertoire :



Sequenced
repertoires

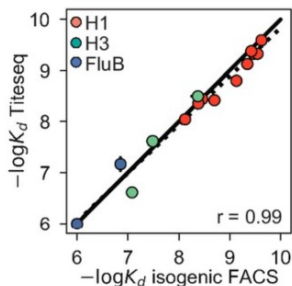


Select
antibodies of
interest



Measure
affinity with
Tite-Seq

Tite-Seq is precise, mature and cheap ^{5 cts / séquence}



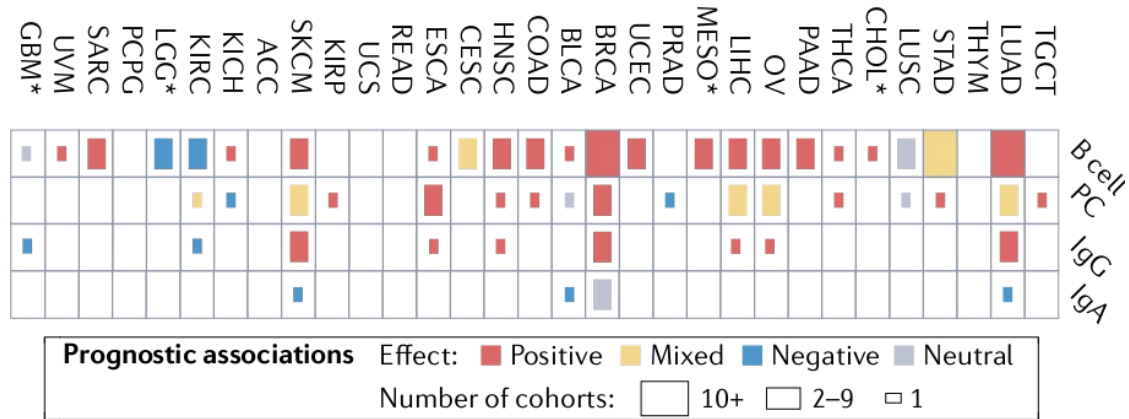
Precision
comparable to
low-throughput
methodes

Starr et al., Nature, 2021
Greaney et al., Cell Host & Microbes, 2023
DeWitt, W. S. et al., Cell, 2026

Tumor Infiltrated B-Lymphocytes (TIL-B)

Tumor Infiltrated B-Lymphocytes (TIL-B)

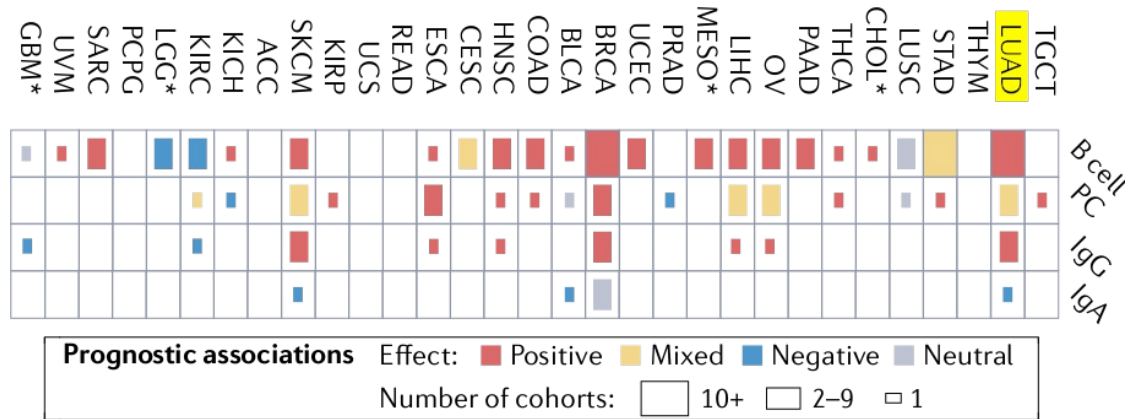
TIL-Bs play a complex role in tumors but generally improve prognosis and response to immunotherapy.



Adapted from *Laumont et al., Nat. Rev. Cancer (2022)*

Tumor Infiltrated B-Lymphocytes (TIL-B)

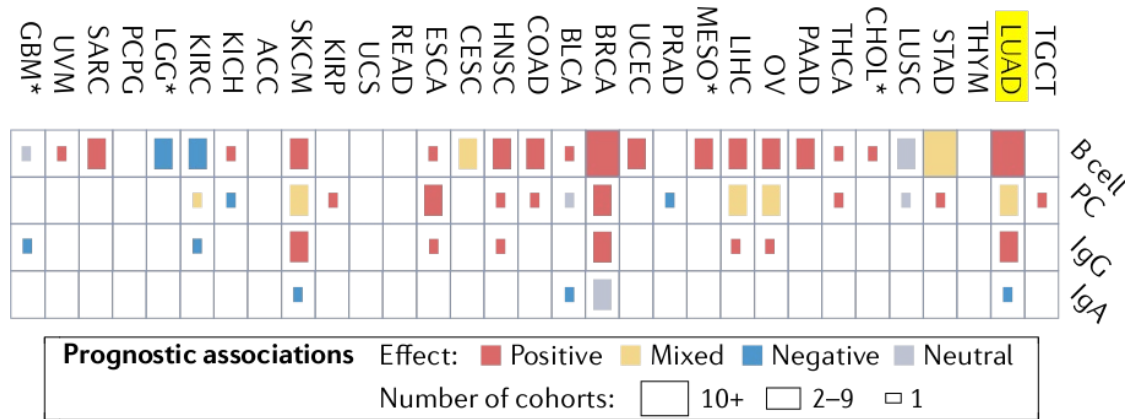
TIL-Bs play a complex role in tumors but generally improve prognosis and response to immunotherapy.



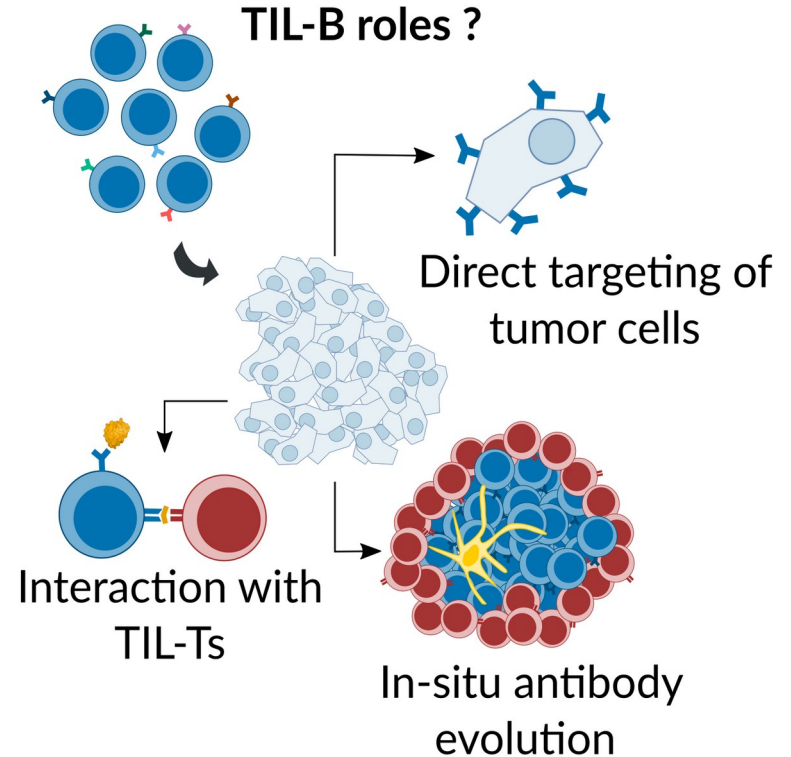
Adapted from *Laumont et al., Nat. Rev. Cancer (2022)*

Tumor Infiltrated B-Lymphocytes (TIL-B)

TIL-Bs play a complex role in tumors but generally improve prognosis and response to immunotherapy.

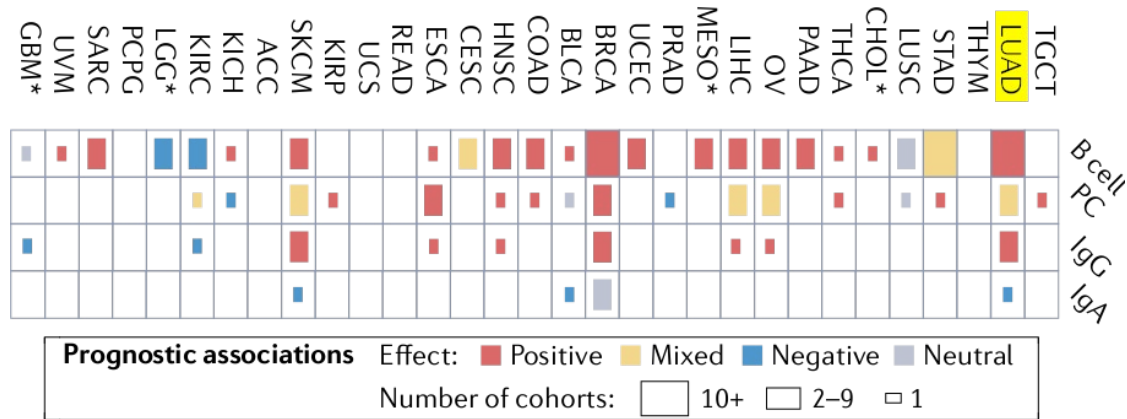


Adapted from *Laumont et al., Nat. Rev. Cancer (2022)*

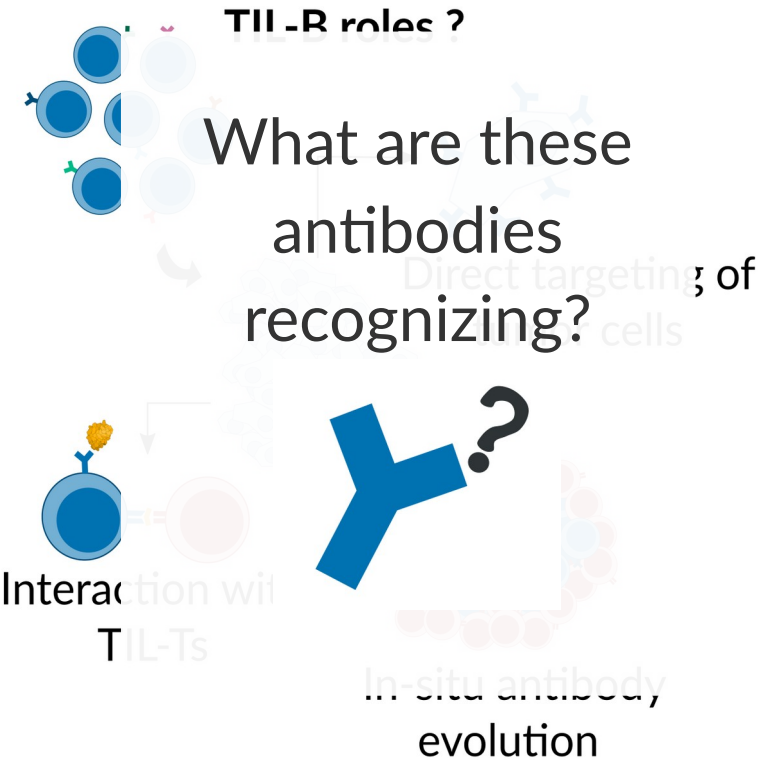


Tumor Infiltrated B-Lymphocytes (TIL-B)

TIL-Bs play a complex role in tumors but generally improve prognosis and response to immunotherapy.



Adapted from Laumont et al., Nat. Rev. Cancer (2022)



The public TIL-B repertoire

Only **70 pairs** antigen /
TIL-antibody identified.

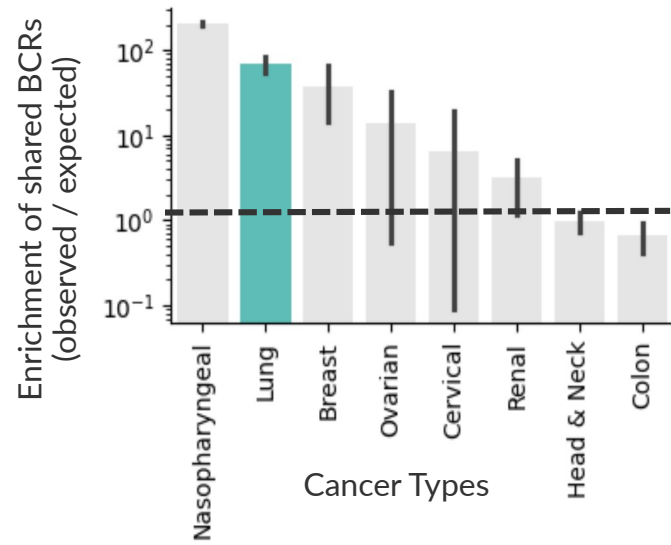
Laumont et al., Nat. Rev. Cancer (2022)

The public TIL-B repertoire

Only 70 pairs antigen /
TIL-antibody identified.

Laumont et al., Nat. Rev. Cancer (2022)

The intra-tumoral repertoire is
public $\Rightarrow$ directed response !



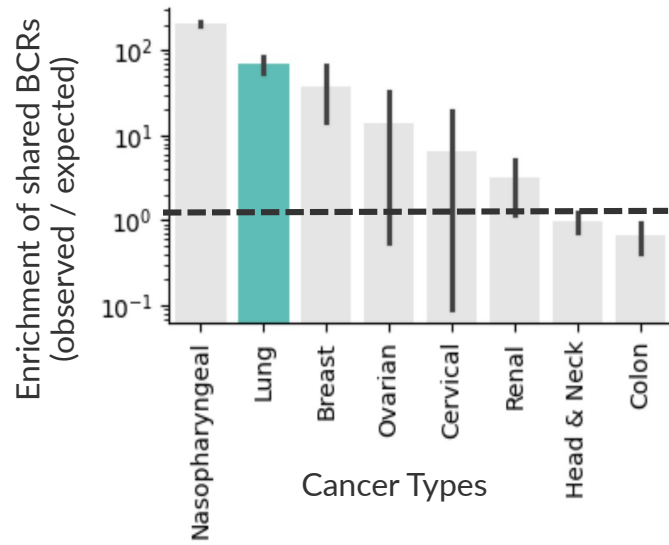
Data from: Aran et al, Front. In Immunol., 2021, McDaniel et al., Cancer Immunol Immunother, 2018, Liu et al., Nat. Commun., 2024, Shu et al., BiorXiv, 2023, Ma et al., Science, 2024, Ruffin et al., Nat. Commun., 2024

The public TIL-B repertoire

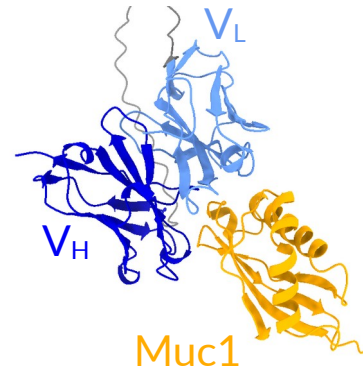
Only **70 pairs** antigen /
TIL-antibody identified.

Laumont et al., Nat. Rev. Cancer (2022)

The intra-tumoral repertoire is
public $\Rightarrow$ directed response !



Data from: Aran et al, *Front. In Immunol.*, 2021, McDaniel et al., *Cancer Immunol Immunother*, 2018, Liu et al., *Nat. Commun.*, 2024, Shu et al., *BiorXiv*, 2023, Ma et al., *Science*, 2024, Ruffin et al., *Nat. Commun.*, 2024



Preliminary Data

Lung cancer:

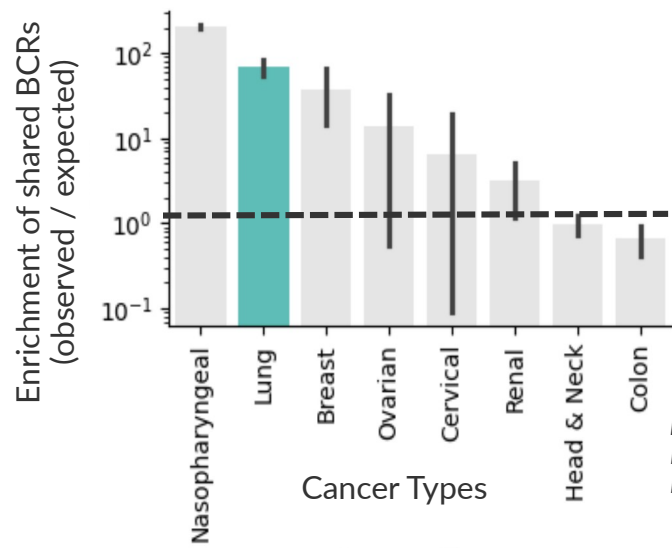
500 BCRs tested
in-silico against 5
antigens

The public TIL-B repertoire

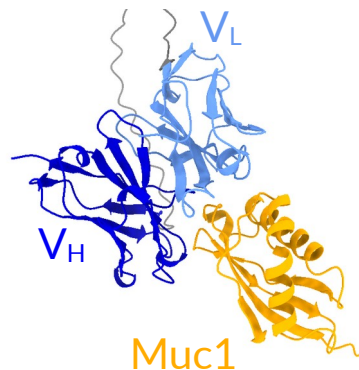
Only **70 pairs** antigen /
TIL-antibody identified.

Laumont et al., Nat. Rev. Cancer (2022)

The intra-tumoral repertoire is
public $\Rightarrow$ directed response !



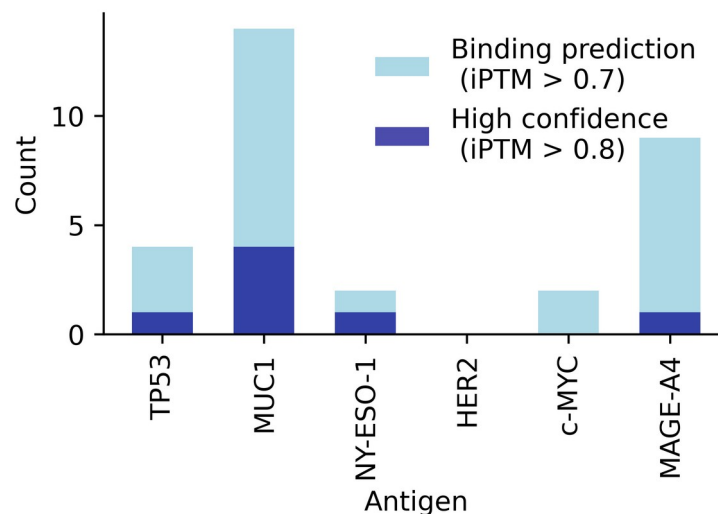
Data from: Aran et al, Front. In Immunol., 2021, McDaniel et al., Cancer Immunol Immunother, 2018, Liu et al., Nat. Commun., 2024, Shu et al., BiorXiv, 2023, Ma et al., Science, 2024, Ruffin et al., Nat. Commun., 2024



Preliminary Data

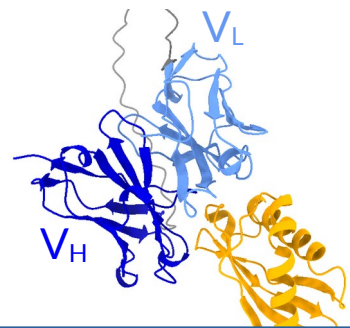
Lung cancer:

500 BCRs tested
in-silico against 5
antigens



The public TIL-B repertoire

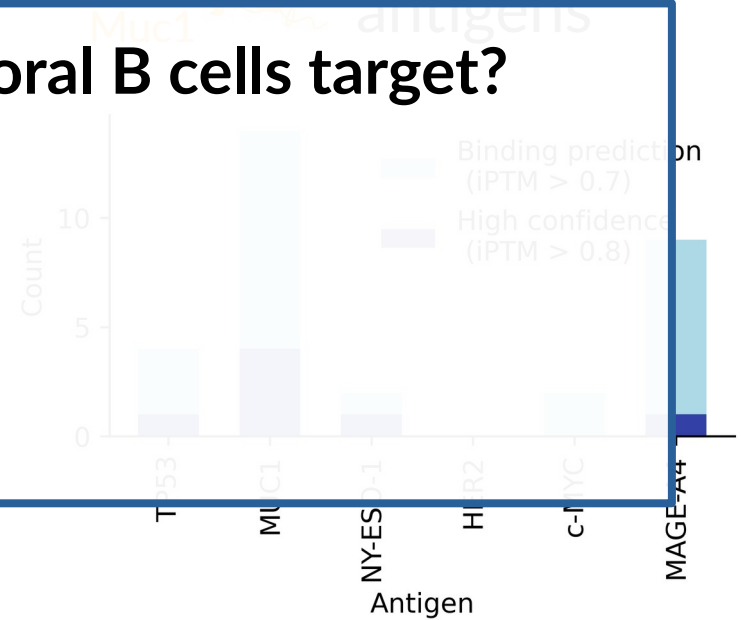
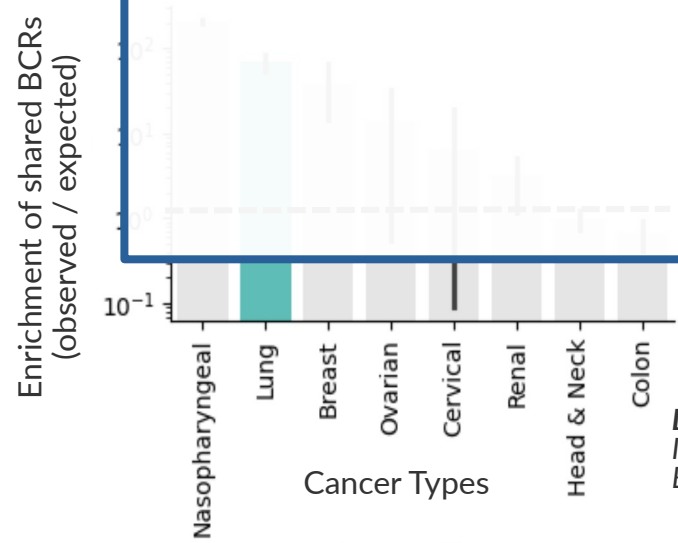
Only 70 pairs antigen / TIL-antibody identified.



Lung cancer:
500 BCRs tested
in-silico against 5

Which antigens do intra-tumoral B cells target?

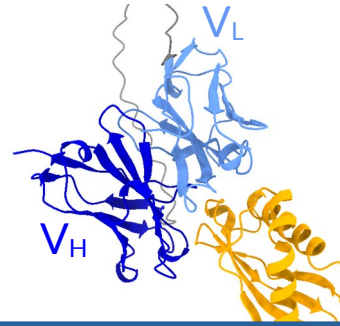
The intra-tumoral repertoire is public $\Rightarrow$ directed response !



Data from: Aran et al, Front. In Immunol., 2021, McDaniel et al., Cancer Immunol Immunother, 2018, Liu et al., Nat. Commun., 2024, Shu et al., BiorXiv, 2023, Ma et al., Science, 2024, Ruffin et al., Nat. Commun., 2024

The public TIL-B repertoire

Only 70 pairs antigen /
TIL-antibody identified.



Preliminary Data

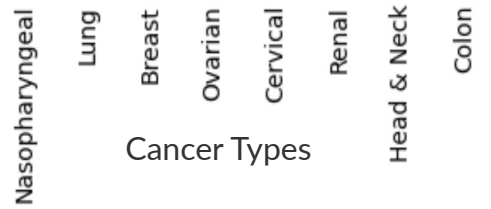
Lung cancer:
500 BCRs tested
in-silico against 5

Which antigens do intra-tumoral B cells target?

- Can the antigens the BCRs target serve as **biomarker**?
- Can we identify novel **targetable** antigens?

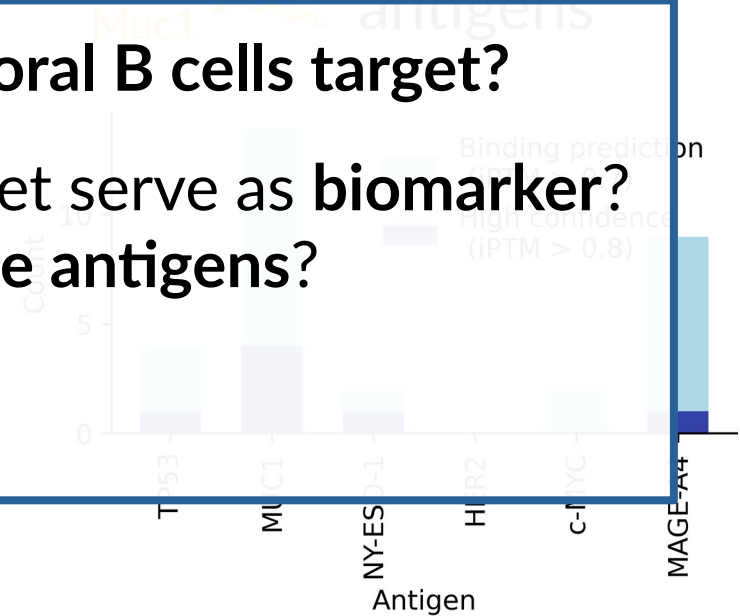
Enrichment of shared BCRs
(observed / expected)

10⁻¹



Cancer Types

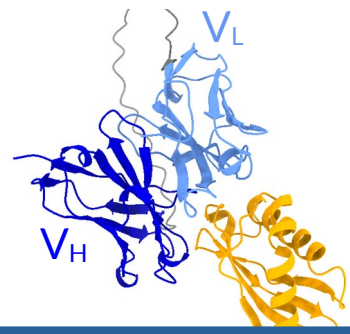
Data from: Aran et al, Front. In Immunol., 2021, McDaniel et al., Cancer Immunol Immunother, 2018, Liu et al., Nat. Commun., 2024, Shu et al., BiorXiv, 2023, Ma et al., Science, 2024, Ruffin et al., Nat. Commun., 2024



Ciml
IMMUNOLOGY

The public TIL-B repertoire

Only 70 pairs antigen / TIL-antibody identified.

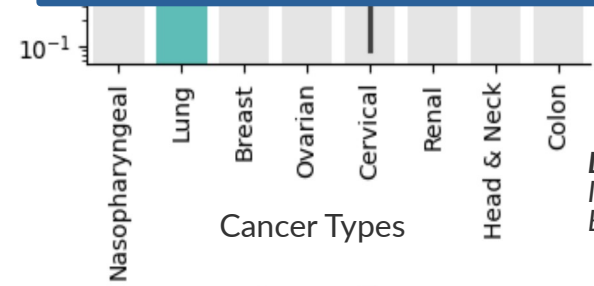


Lung cancer:
500 BCRs tested
in-silico against 5

Which antigens do intra-tumoral B cells target?

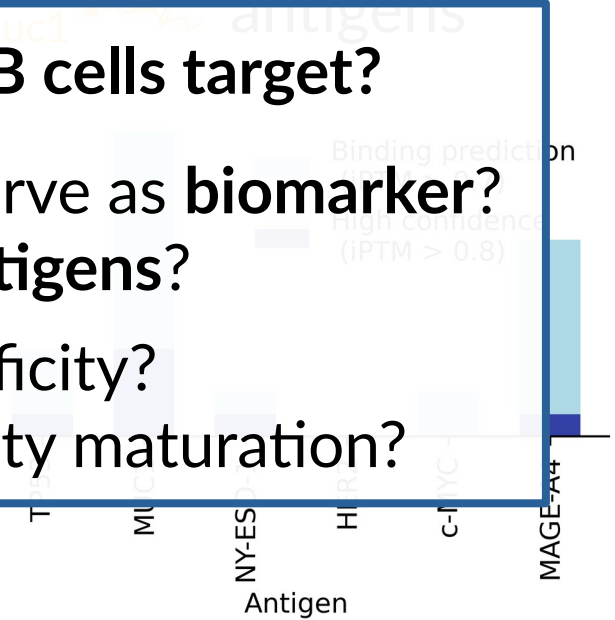
- Can the antigens the BCRs target serve as **biomarker**?
- Can we identify novel **targetable** antigens?
- Do plasma cells have different specificity?
- Does specificity change during affinity maturation?

Enrichment of shared BCRs
(observed / expected)



Cancer Types

Data from: Aran et al, Front. In Immunol., 2021, McDaniel et al., Cancer Immunol Immunother, 2018, Liu et al., Nat. Commun., 2024, Shu et al., BiorXiv, 2023, Ma et al., Science, 2024, Ruffin et al., Nat. Commun., 2024

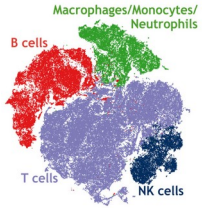


Antigen

Which antigens are targeted by TIL-Bs



Non-small cell lung
cancer dataset(s)



Public data

Hao (2022)

Prazanowska (2023)

...

RHU
LUCA-PI

900 prospective patients
(pulmonary nodules,
PBMC)

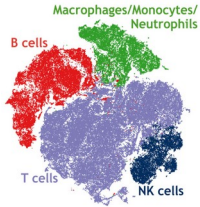


216 patients
(tumor & PBMC
samples)

Which antigens are targeted by TIL-Bs



Non-small cell lung cancer dataset(s)



Public data

Hao (2022)

Prazanowska (2023)

...

RHU
LUCA-PI

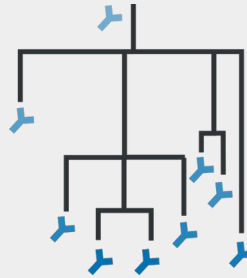
900 prospective patients
(pulmonary nodules,
PBMC)



216 patients
(tumor & PBMC
samples)

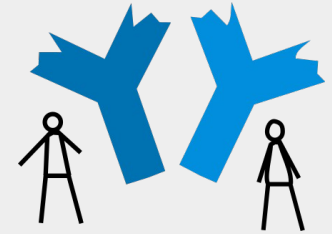


Identify TIL-B of interest *in-silico*
(WP2)



Clonality, isotypes &
transcriptomics

Plasma / Memory B-cells.
Clonal lineages under
selection



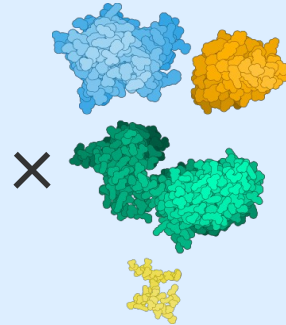
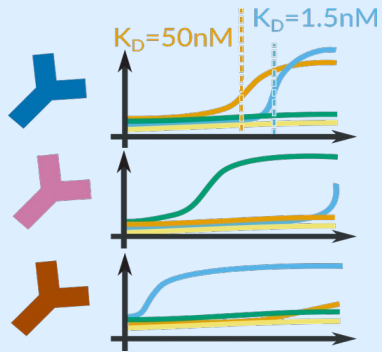
Comparisons
between patients

Identify similar receptors
between repertoires

Which antigens are targeted by TIL-Bs



Measure the specificity of the repertoire *ex-vivo* (WP1/WP3)



Neoantigens
(EGFR L858R,
EML4-ALK...)

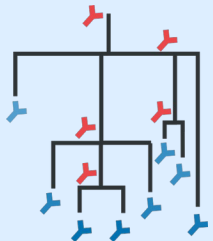
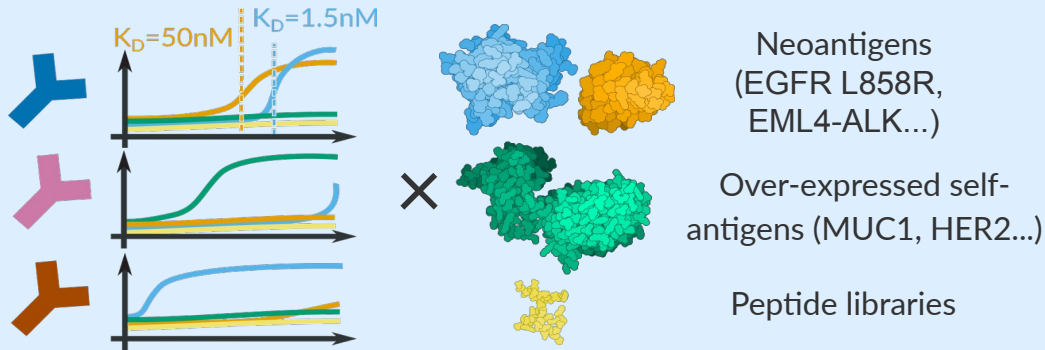
Over-expressed self-
antigens (MUC1, HER2...)

Peptide libraries

Which antigens are targeted by TIL-Bs



Measure the specificity of the repertoire *ex-vivo* (WP1/WP3)



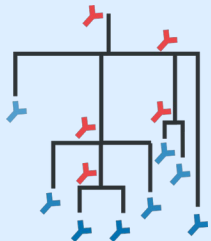
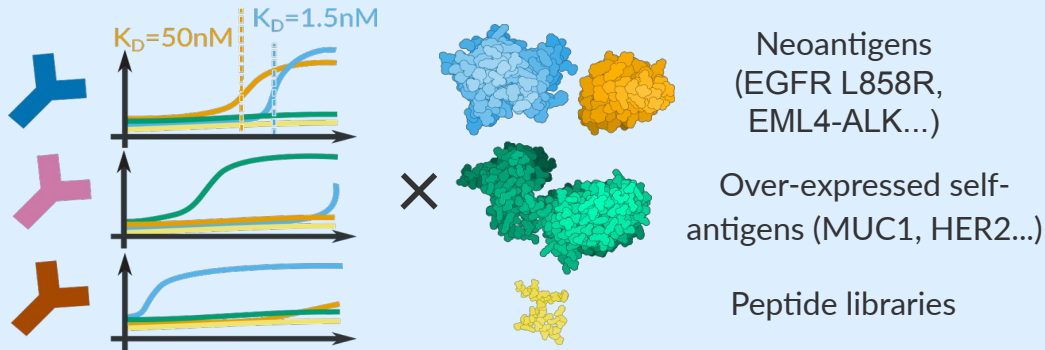
Ancestral
reconstruction

Did the antibody evolve
against this antigen ?

Which antigens are targeted by TIL-Bs



Measure the specificity of the repertoire *ex-vivo* (WP1/WP3)



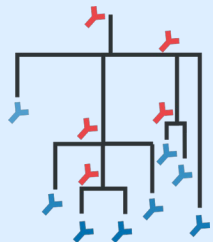
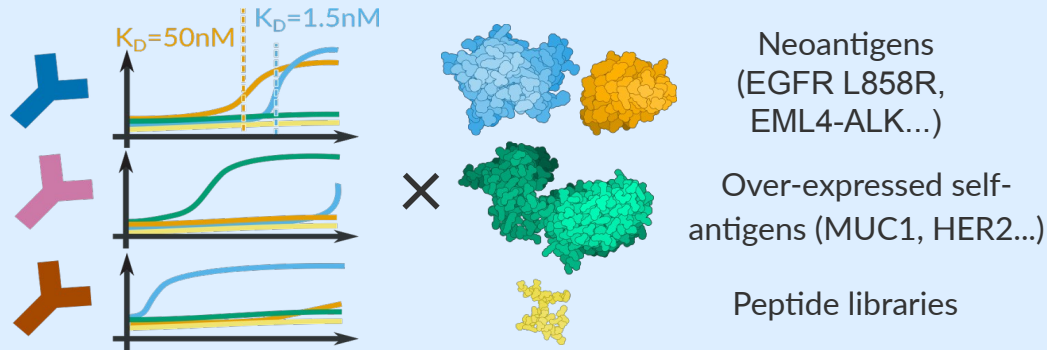
Ancestral reconstruction
Did the antibody evolve against this antigen?

Comparison with sera / PBMC
Pioneer / LUCA-PI Cohort
PhiP-Seq

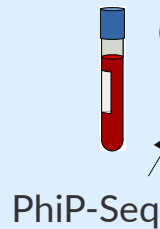
Which antigens are targeted by TIL-Bs



Measure the specificity of the repertoire *ex-vivo* (WP1/WP3)



Ancestral reconstruction
Did the antibody evolve against this antigen ?

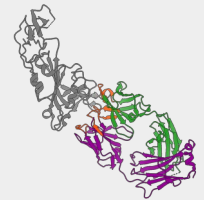


Comparison with sera / PBMC
Pioneer / LUCA-PI Cohort



Models & learning (WP2)

Generalize Ab-Ag predictions



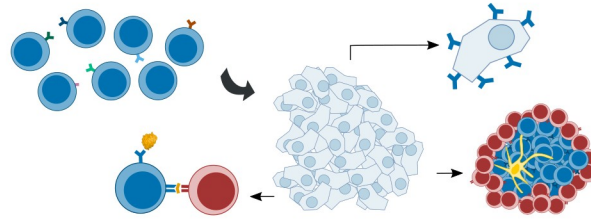
Integrate specificity to prognosis
A person icon followed by a large question mark.

THOMAS DUPIC

Affinity landscape of intra-tumoral B-cells in lung cancer

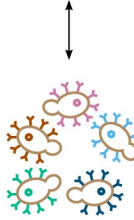


Project summary



Which antigens do intra-tumoral B cells target?

- BCRs as biomarkers
- New actionable antigens ?
- Plasma cell specificity
- Evolution of specificity

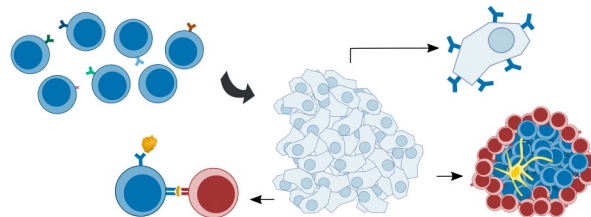


THOMAS DUPIC

Affinity landscape of intra-tumoral B-cells in lung cancer

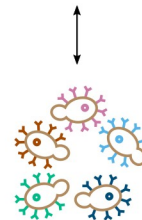


Project summary

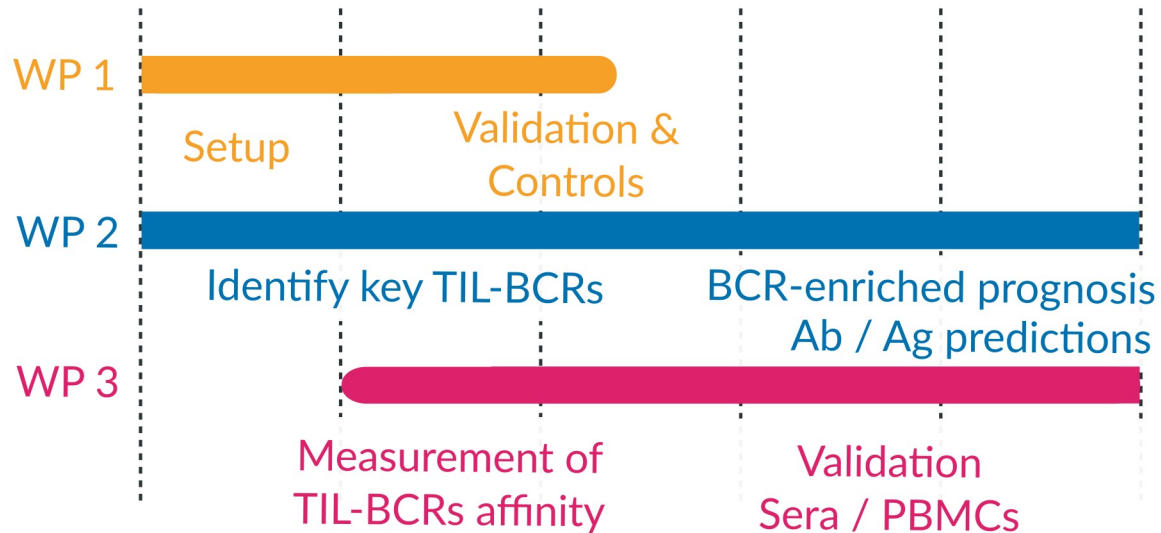


Which antigens do intra-tumoral B cells target?

- BCRs as biomarkers
- New actionable antigens ?
- Plasma cell specificity
- Evolution of specificity



Timeline



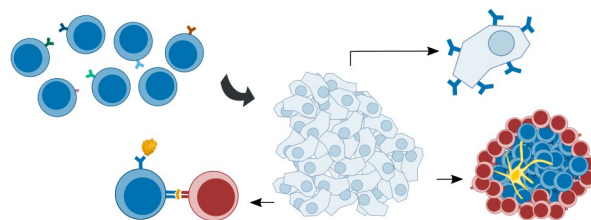
Post-doctoral researcher

THOMAS DUPIC

Affinity landscape of intra-tumoral B-cells in lung cancer

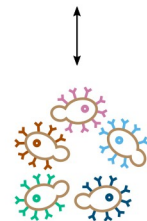


Project summary



Which antigens do intra-tumoral B cells target?

- BCRs as biomarkers
- New actionable antigens ?
- Plasma cell specificity
- Evolution of specificity



Partner institutions

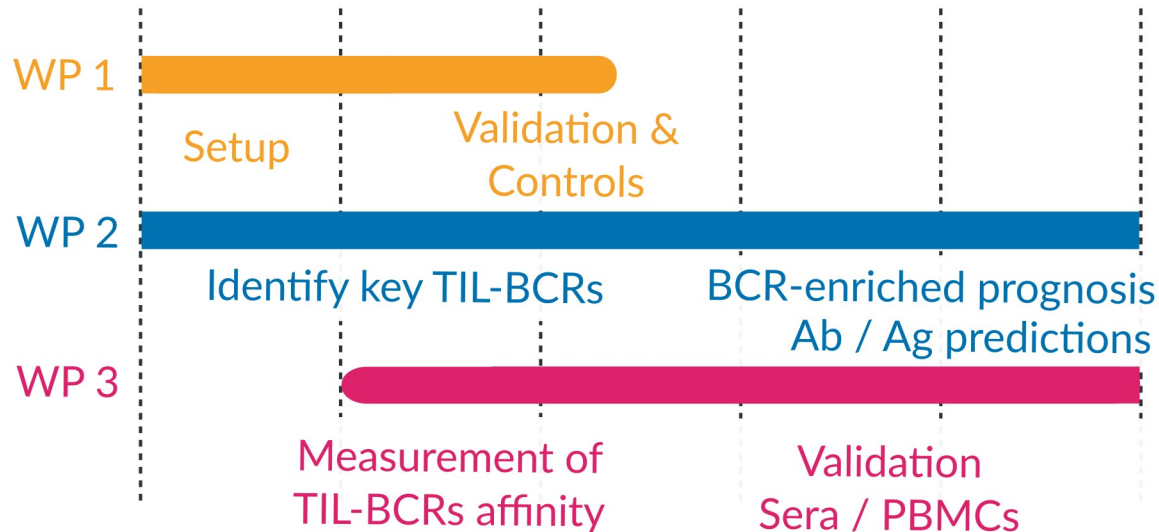


Co-financing
Idex $\Rightarrow$ 1 PhD



Team leader
Equipment,
lab, technician

Timeline



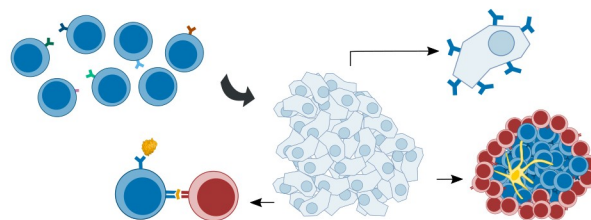
Post-doctoral researcher

THOMAS DUPIC

Affinity landscape of intra-tumoral B-cells in lung cancer

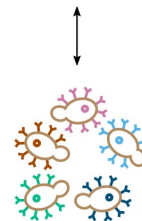


Project summary

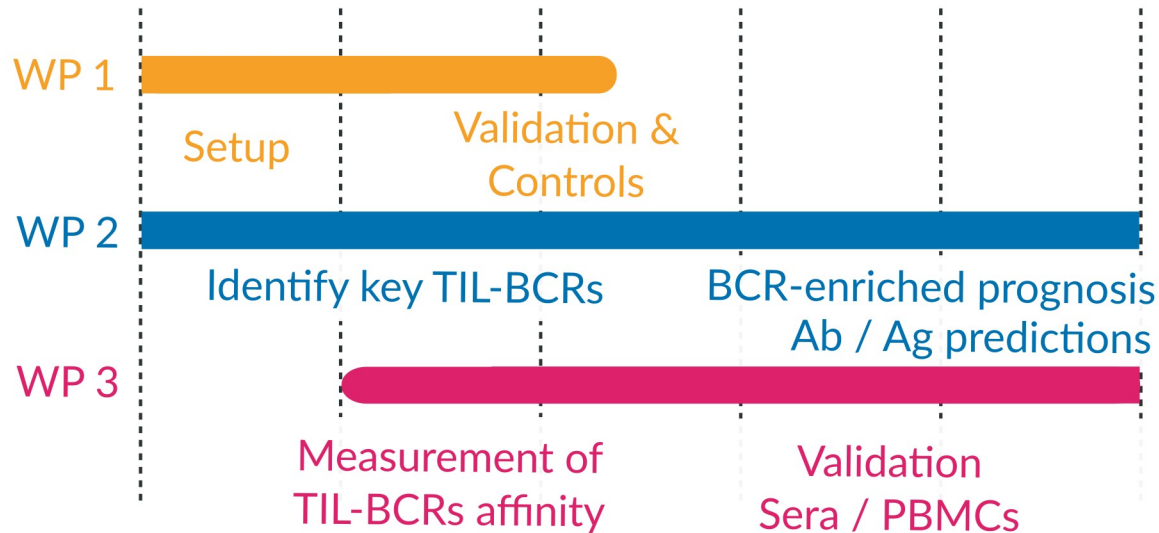


Which antigens do intra-tumoral B cells target?

- BCRs as biomarkers
- New actionable antigens ?
- Plasma cell specificity
- Evolution of specificity



Timeline



Post-doctoral researcher

Partner institutions

Inserm

amU
Aix Marseille Université

Co-financing
Idex $\Rightarrow$ 1 PhD

ap. hm

Cimi
IMMUNOLOGY

Team leader
Equipment,
lab, technician

Collaborations

CRCM
Centre de Recherche
en Cancérologie de Marseille

UCSF
 LPENS
LABORATOIRE DE PHYSIQUE
DE L'ÉCOLE NORMALE SUPÉRIEURE

P. Milpied
Q. Marcou
A. Phillips
T. Mora, A. Walczak

LUCA-PI

THE PIONEER PROJECT